

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

POLYMER SCIENCE
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define the term elastomers with respect to Tg	M1.01	R
2	Why polymers have average molecular weight?	M2.01	U
3	Suggest a method to find Mn of polymer?	M2.04	U
4	Define polymer composites	M4.05	R
5	Define the term melting temperature	M1.02	R
6	Expand the terms TGA and DSC	M1.04	R
7	What are stereoregular polymers	M3.03	U
8	What is grafting	M4.05	U
9	Define tacticity	M3.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Distinguish between amorphous and crystalline polymers with example	M1.03	U
2	Derive the expression for Mw*	M2.03	U
3	Write short note on polydispersity and molecular weight distribution	M2.02	U
4	Distinguish between isotactic, syndiotactic and atactic polymers	M3.02	U
5	Write short note micro structure based on chemical structure.	M3.01	U
6	Draw the chemical structures of cis -1,4polyisoprene, trans -1,4polyisoprene and 1,2 polyisoprene.	M3.04	U
7	Discuss the effect of aldehydic groups in polymers	M4.02	U
8	Explain the importance of Z-N catalyst in stereo regularity	M3.03	U
9	Discuss any three factors affecting Tg of polymers with examples.	M1.03	U

10	Define crystallizability. Write one factor that can affect the crystallizability with example.	M1.03	U
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PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain the analysis of polymers using TGA and DSC	M1.04	U
OR			
2	Explain the method of determining Tg by dilatometry	M1.03	U
3	Explain gel permeation chromatography for the determination of molecular weight	M2.04	U
OR			
4	A sample of high polymer contains 20% by weight of macromolecules of molecular weight 10000 and 80% by weight of macromolecules of molecular weight 100000. Calculate the value of Mn and Mw.	M2.03	A
5	Explain the techniques of determination of Mn by end group analysis and cryoscopy.	M2.04	U
OR			
6	Explain the determination of molecular weight using viscometry	M2.04	U
7	Discuss the possible microstructures of BR and IR	M3.04	U
OR			
8	Explain geometrical and optical isomerism in polymers with examples	M3.02	U
9	Discuss following polymer reactions of (a) Hydrolysis (b) acydolysis	M4.03	U
OR			
10	Explain polymer modification by crosslinking and Interpenetrating network	M4.04	U
11	Explain the effect of hydroxyl and carboxylic groups in a polymer	M4.02	U
OR			
12	Explain physical methods for polymer modification	M4.04	U

Prepared By:

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Lecturer

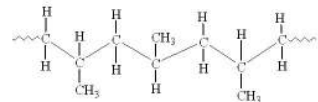
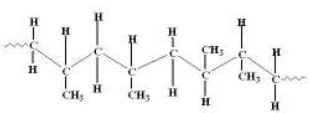
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GPTC Adoor

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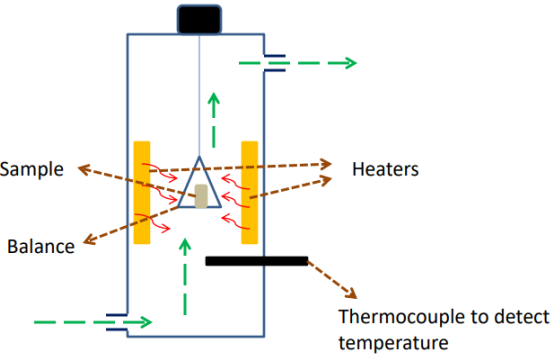
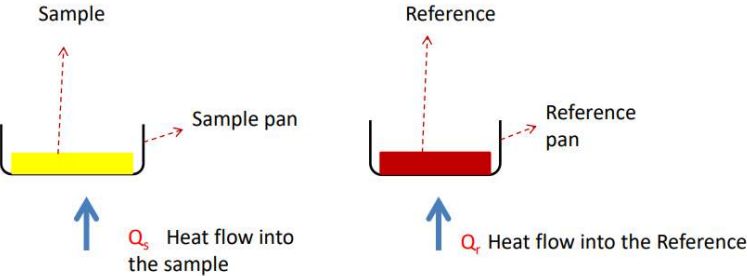
SCHEME OF VALUATION**(Scoring Indicators)**

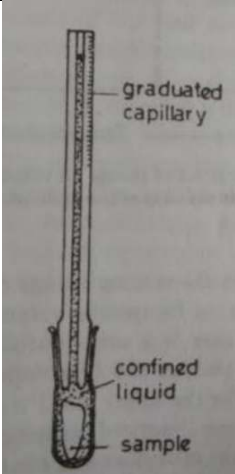
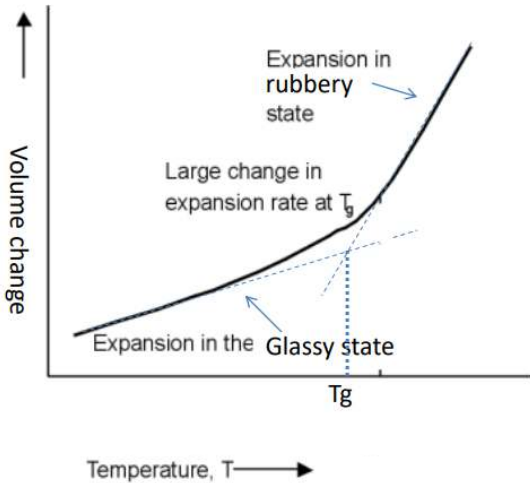
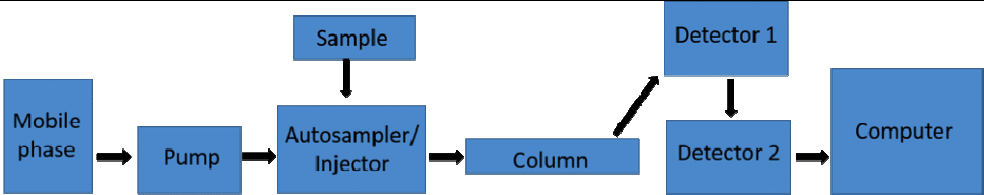
Revision : 2021		Course Code :		
Course Title :Polymer science - Set 1				
Qst. No	Scoring Indicator	Split up Score	Sub Total	Total
	PART A			
I. 1.	Elastomers are polymers with Tg lower than room temperature	1	1	1
2.	Polymers are polydispersed, so they don't have a single molecular weight	1	1	1
3.	Ebulliometry, Cryoscopy, Osmometry, end group analysis	1	1	1
4.	Composite material with a polymer matrix	1	1	1
5.	Temperature at which crystalline domains of polymer melt into liquid	1	1	1
6.	Thermo gravimetric analysis, Differential scanning calorimetry	0.5*2	1	1
7.	Polymers whose linear molecules consists of monomeric units having either identical spatial configurations or configurations that, while not identical, alternate in regular fashion.	1	1	1
8.	Graft polymers are segmented copolymers with a linear backbone of one composite and randomly distributed branches of another composite	1	1	1
9.	Transition metal halides and alkyl or aryl compounds of group 1 to 4	1	1	1
	PART B			
1.	Amorphous polymer - No ordered arrangement and close packing of atoms or molecules, Transparent, No crystalline melting point Shows clear crystalline melting point, Clear glass transition is observed, Example :- Polycarbonate, Polystyrene, Polymethyl methacrylate Crystalline polymer - Ordered arrangement and close packing of atoms or molecules are observed, Opaque, Hard to observe glass transition or it is dependent on the amount of crystallinity Example :- Polyethylene, Polypropylene, Nylon 6, Nylon 66, Polyacetal	1.5 1.5	3	3
2.	Let there be N1 number of polymer molecules with M1 molecular weight N2 number of polymer molecules with M2 molecular weight N3 number of polymer molecules with M3 molecular weight Ect... Ni number of polymer molecules with Mi molecular weight Total number of polymer molecules = N1 +N2 +N3 +....+Ni = Σ Ni Number fraction of N1 molecules = N1 /Σ Ni Number fraction of N2 molecules = N2/ Σ N2 Ect...	3	3	3

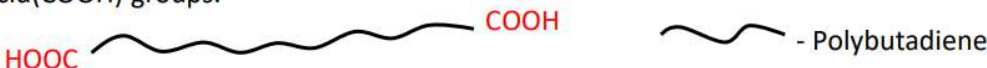
	Number fraction of Ni molecules = $N_i / \sum N_i$ Contribution of polymer molecule with M1 molecular weight on number average molecular weight of polymer sample = Number fraction of M1 molecule * M1 $N_1 * M_1 / \sum N_i$ Contribution of polymer molecule with M2 molecular weight on number average molecular weight of polymer sample = $N_2 M_2 / \sum N_i$ Contribution of polymer molecule with Mi molecular weight on number average molecular weight of polymer sample = $N_i M_i / \sum N_i$ Number average molecular weight of polymer sample = sum of the contribution from polymer chains with different molecular weights = $N_1 M_1 / \sum N_i + N_2 M_2 / \sum N_i + N_3 M_3 / \sum N_i + \dots + N_i M_i / \sum N_i = \sum N_i M_i / \sum N_i$			
3.	Polymer molecules in a polymer sample will not be identical; it will be a mixture of molecules with different molecular weights. This is called polydispersity. It is expressed in terms of Polydispersity index (PDI). $PDI = M_w/M_n$. Molecular weight distribution of a polymer sample can better studied from number of molecules Vs Molecular weight plot. If the graph is broader then polymer is having a broader molecular weight distribution. Polymers with broader molecular weights can be processed easily but their properties are not good as polymers with narrow molecular weight distribution.	1.5	3	3
4.	Tacticity is observed in polymers with repeating units that contain asymmetric carbon atom. If the groups attached to asymmetric carbon is on the same side of polymer backbone then such polymers are isotactic. If the groups are arranged in alternative fashion then such polymers are called syndiotactic. If there is no order then such polymers are called atactic. Isotactic  Syndiotactic  Atactic 	3	3	3
5.	Microstructure of the polymer can be studied based on chemical structure or chemical composition. Basic chemical structure and composition determines the properties of polymers. Based on the chemical structure polymers are classified as ☐ Organic and inorganic polymers - Organic polymers are polymers which contain carbon atoms in the polymer backbone. Eg:- polyethylene, polypropylene, PS ect... Inorganic polymers are polymers with carbon free backbone. Eg:- Silicone	3	3	3

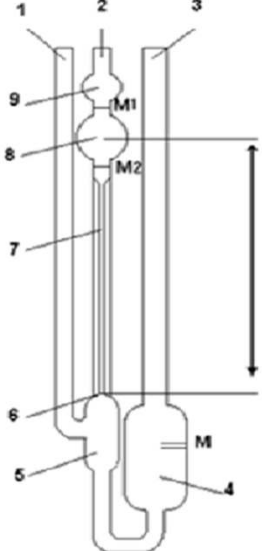
	polymers, Polygermaneeect.. □ Homo chain and hetero chain polymers – Polymer backbone containing only one type of atom is called homo chain polymers eg:- PE and others called hetero chain polymers eg:- nylon. □ Homo polymers and copolymers – Polymers made from same monomers are called homopolymerseg:- PP, Polymers with more than one type of repeaing units are called copolymers eg:- SBR.			
6.	$\begin{array}{c} \text{---CH}_2 \quad \text{CH}_2\text{---} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{H} \end{array}$ Cis 1,4 Poly(isoprene) $\begin{array}{c} \text{---CH}_2 \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{CH}_2\text{---} \end{array}$ trans 1,4 Poly(isoprene) $\begin{array}{c} \text{CH}_3 \\ \\ \left[\text{CH}_2 - \text{C} \right]_n \\ \\ \text{CH} \\ \\ \text{CH}_2 \end{array}$ 1,2 polyisoprene	1*3	3	3
7.	Some of the polymers contain aldehyde groups. These carboxylic acid groups can undergo reactions shown by simple aldehydes.Example for the polymer containing aldehyde group is polyacrolein $\left\{ \text{CH}_2 - \underset{\text{CHO}}{\underset{ }{\text{CH}}} \right\}_n$ Aldehyde group in polyacrolein can react with sodium bi sulfate. This reaction can convert polyacrolein from a water insoluble polymer into a water soluble polymer. $\begin{array}{c} \sim\text{CH}_2 - \underset{\text{O}=\text{C}}{\underset{\text{H}}{\underset{ }{\text{CH}}}}\sim \\ + \\ \text{NaHSO}_3 \\ \downarrow \\ \sim\text{CH}_2 - \underset{\text{HO}-\text{C}-\text{SO}_3\text{Na}}{\underset{\text{H}}{\underset{ }{\text{CH}}}}\sim \end{array}$ Polyacrolein can react with hydroxyl amine	3	3	3

8.	Zeigler natta catalysts are coordination catalysts. Zeigler natta catalyst consists of two components 1) Halides of transition metals (Catalyst), 2) Alkyls, aryls and hydrides of group 1 to 4 elements (Co catalyst). They are widely used for the production of stereo regular polymers. A polymer with a repeating unit that contains an asymmetric carbon atom can have different special arrangement for repeating units. Zeigler natta catalyst can be used to obtain a polymer with regular special arrangement or stereo regular. Polymers with regular structures show unique properties. Stereo regular polypropylene (isotactic, syndiotactic) can crystallize easily and they show good mechanical and thermal properties. This makes them very useful. Stereo irregular PP is not a useful product.	3	3	3
9.	1. Flexibility of the bonds in polymer backbone – Polymers with flexible back bone are having low Tg. Example:- silicone rubber is having very flexible backbone and its Tg is very low. 2. Presence of aromatic/cyclic groups in polymer backbone – Polymers with aromatic or cyclic groups in their backbone show higher Tg. These groups reduce the polymer chain flexibility. Example:- PET is having higher Tg than PE. 3. Bulkiness of side groups- Bulky side groups also reduces the polymer chain flexibility and such polymers show higher Tg. Example PS having higher Tg than PE. 3. Intermolecular forces – Intermolecular forces restrict polymer chain movement. So polymers with high inter molecular forces show higher Tg. Example PVC is having higher Tg than PE 4. Crosslinking - Crosslinking increases the glass transition temperature of polymers. Example crosslinking of natural rubber with excess of sulphur results in ebonite. 5. Copolymerization – Copolymer will show a Tg in between the Tg's of homopolymers made up of monomers used for its preparation. Example SBR made from styrene and butadiene. 6. Incorporation of plasticizers – Incorporation of plasticizer reduces the glass transition temperature of polymer. Example plasticized PVC.	1*3	3	3
10.	Crystallisability is the maximum crystallinity that a polymer chain can achieve at a particular temperature regardless of other conditions of crystallization. Factors affecting crystallizability •Geometric regularity – Linear polymers can crystallize easily compared to branched polymer because linear polymers are having more regular structure. Example linear PE (HDPE) crystallize better than branched PE (LDPE) •Stereo regularity/Tacticity – If the polymer is more stereo regular then it can crystallize easily. Isotactic and syndiotactic polymers can crystallize easily because they are having structural regularity. •Macromolecular geometry – macro molecular geometry can affect crystallization. NR	1 1*2	3	3

	molecules having a coiled structure and they can crystallize upon cooling or stretching. Gutta-purca molecules are chemically identical but having a rod like structure. They cannot crystallize •Copolymerization – Copolymerizaion produce polymer with less structural regularity. So copolymerizaion reduce crysallizability. Example polyethylene and polypropylene are crystalline bu heir copolymer is amorphous. •Intermolecular forces - Inter molecular forces enhances close packing of polymer chains. So intermolecular forces favour crystallization. •Bulky side groups - Bulky side groups reduces polymer chain flexibility and makes chain packing difficult. So hey reduce crystallizability.			
III				
1.	 • Sample is placed on the balance before starting the experiment. • Sample is heated at a specific rate (Eg: - 1 °C per minute). • A gas (eg:- nitrogen, air, oxygen ect...) is passes through the sample chamber throughout the experiment. This helps in the removal of volatile materials evolving from sample. • Weight of the sample is measured as a function of temperature. • Weight of the sample is plotted against temperature. • When sample degrades a reduction in weight is observed. • From this the degradation temperature of a polymer can be found • Degradation temperature is different for different polymers  •Sample and reference are heated/cooled according to predetermined temperature programme. Difference between Qs and Qr is measured at each temperature • Difference in heat flow is plotted against temperature • From his graph extract Tg,Tm, Tcect... • By finding Tg and Tm (if it is a crystalline polymer) of the polymer sample it is	3.5	7	7
		3.5		

	possible to predict the polymer			
OR				
2.	 • Fill the bulb of dilatometer with sample and an inert liquid. • Fix the top segment with capillary as shown in figure. • Volume of the sample changes when it is heated or cooled. • Volume change of the sample can be calculated from the level of liquid in capillary. • Volume change of the sample with temperature is noted. • Volume change/specific volume values are plotted against temperature • Slope of graph changes at glass transition temperature. 	7	7	7
3.	 Gel permeation chromatography is a technique used to separates molecules in a mobile phase or solvent according to their size. Main part of the equipment is a column filled with porous beads. Porous beads were made using polystyrene divinyl benzene copolymers. Once a mixture of molecules with different sizes are passed through this column the smaller molecules gets trapped in pores and biggest molecules come out easily. Time taken for a sample/sample fraction to come out of the column is inversely	7	7	7

	proportional to its molecular weight. Column is calibrated using standard samples of different molecular weight to quantify this relationship. To test a sample dissolves the polymer in a suitable solvent. Once dissolved in appropriate solvent, inject sample into continuously flowing mobile phase of the same solvent. Dissolved sample and the mobile phase are then pumped through a column. Sample molecules separate according to their size. This finally leads to a detector and the data system.			
OR				
4.	$M_1 = 10000$ $M_2 = 100000$ Let the weight of sample be 1000000 Then $M_1N_1 = 200000$ $M_2N_2 = 800000$ $N_1 = 200000/10000 = 20$ $N_2 = 800000/100000 = 8$ $N_1M_1^2 = N_1M_1 * M_1 = 200000 * 10000 = 2000000000$ $N_2M_2^2 = N_2M_2 * M_2 = 800000 * 100000 = 80000000000$ $M_n = (M_1N_1 + M_2N_2)/(N_1 + N_2) = (200000 + 800000)/20 + 8 = 35714$ $M_w = \frac{(N_1M_1^2 + N_2M_2^2)}{(M_1N_1 + M_2N_2)} = \frac{(2000000000 + 80000000000)}{(200000 + 800000)} = 820000$	7	7	7
5.	End group analysis Useful to find the molecular weight of polymers having functional in their chain ends. In order to determine molecular weight molecule should contain reactive end groups. Number/concentration of end groups in a polymer sample can be found by titration method. Number average molecular weight can be found from titration results. This technique can be used for the determination of molecular weight of polymers like Carboxyl Terminated Polybutadiene (CTPB), Hydroxyl Terminated Polybutadiene (HTPB), Polyethylene glycol ect... Example: Determination of the molecular weight of carboxyl terminated polybutadiene. Here every polymer chain contains 2 carboxylic acid(COOH) groups.  HOOC ————— COOH ————— - Polybutadiene - Known amount CTPB is dissolved in toluene –ethanol mixture. - It is titrated against standardized KOH solution in alcohol. - Carboxyl equivalent = $\frac{V * N}{W * 1000}$ V – volume of KOH solution used N – Normality of KOH solution W – weight of sample - Number average molecular weight ($\bar{M}_n$) = $\frac{2}{\text{Carboxyl equivalent}}$ Cryoscopy	3.5	7	7

	Freezing point of a liquid/solvent reduces if a solute is added into it. Depression in freezing point of the solvent depends on amount or thenumber of solute molecules present in it. Determination of polymer molecular weight using cryoscopy is based on this principle. Number average molecular weight is measured using this method. Relation between molecular weight of polymer sample and itsconcentration in solution is shown in the equation $\frac{\Delta T_f}{C} = \frac{RT_f^2}{\rho \Delta H_f} \left(\frac{1}{M_n} + BC \right)$ C – concentration of polymer in the solution ΔT_f - Difference between the freezing points of pure solvent and the solution T_f - Freezing point of pure solvent R – Universal gas constant ΔH_f - Heat of fusion of solvent ρ – Density of solvent B - Constant Freezing points of polymer solutions with different polymerconcentrations are measured. Calculated the difference between the freezing point of pure solvent and the solution at each case.Divided the difference with concentration and plotted it against concentration. Molecular weight is calculated from the values obtained from this graph.	3.5		
OR				
6.	 1. Ventilation tube 2. Capillary tube 3. Filling tube 4. Lower reservoir 5. Suspended level tube 6. Capillary 7. Timing Bulb 8. Upper reservoir 9. M1 and M2 Measuring marks M filling marks h mean driving head Fig. 1 Ubbelohde viscometer ● Solvent is filled into the reservoir M and transferred into upper reservoir using a rubber bulb. ● Rubber bulb is removed so that the liquid starts flowing through the capillary from top to bottom. ● Calculate the time required to lower the liquid level from M1 to M2 using a stop watch.	7	7	7

- Repeat the same for polymer solutions with different concentrations.
- Reduced viscosity value and inherent viscosity value for each concentration are calculated and plotted against concentration
- Intrinsic viscosity value is calculate from the graph
- Molecular weight is calculated using **Mark – Houwink equation**

$$\frac{t}{t_0} = \frac{\eta}{\eta_0}$$

t – Flow time of polymer solution
 t_0 – Flow time of solvent
 η – Viscosity of polymer solution
 η_0 – Viscosity of solvent

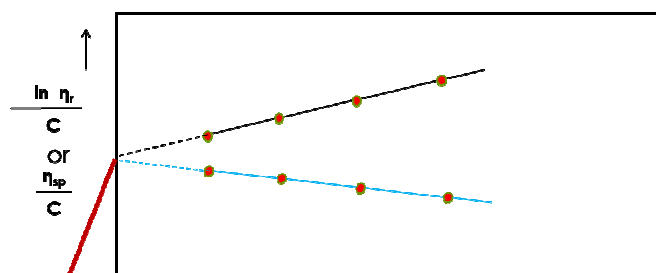
$$\frac{\eta}{\eta_0} = \frac{t}{t_0} = \text{Relative viscosity } (\eta_r)$$

$$\eta_r - 1 = \text{Specific viscosity } (\eta_{sp})$$

C – concentration of solution in g/dl

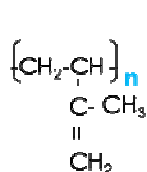
$$\frac{\eta_{sp}}{C} = \text{Reduced viscosity } (\eta_{red}) \text{ (dl/g)}$$

$$\frac{\ln \eta_r}{C} = \text{Inherent viscosity } (\eta_{inh}) \text{ (dl/g)}$$



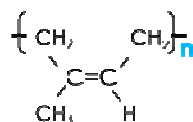
$$\left(\frac{\ln \eta_r}{C} \right)_{C \rightarrow 0} = \left(\frac{\eta_{sp}}{C} \right)_{C \rightarrow 0} = \text{Intrinsic viscosity } ([\eta])$$

7. Different struces and spacial arrangemens are possible for the repeating unit of IR and BR depending on the reaction of monomers and their spcial arrangement durin polymerizaion. Possible srucures of BR an IR are given below.



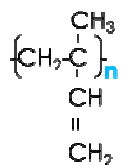
3,4 poly(isoprene)

3,4 addition



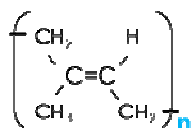
Cis 1,4 Poly(isoprene)

1,4 addition



1,2 poly(isoprene)

1,2 addition

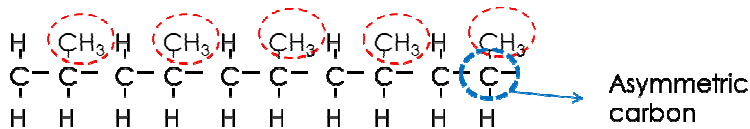
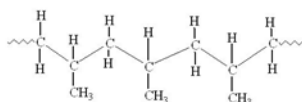
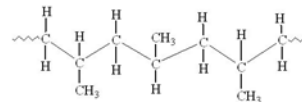
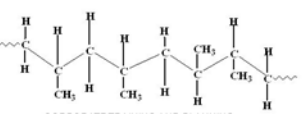
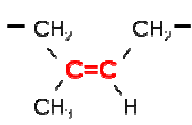
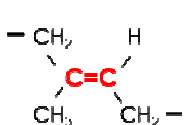


trans 1,4 Poly(isoprene)

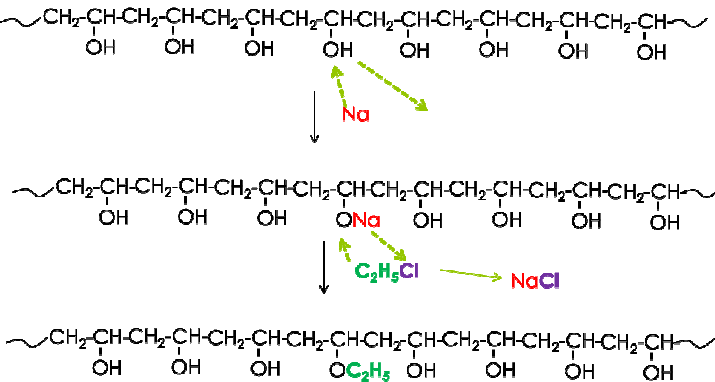
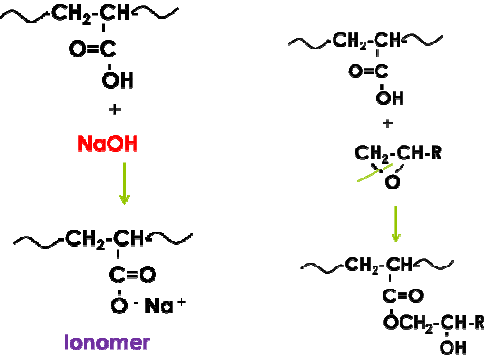
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	$\left[\begin{array}{c} \text{CH}_2 - \text{CH} \\ \\ \text{C} - \text{H} \\ \\ \text{CH}_2 \end{array} \right]_n$ 1,2 poly(buadiene) $\left[\begin{array}{c} \text{CH}_2 \quad \text{C} - \text{H} \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \quad \end{array} \right]_n$ 1,4 Poly(cutadiene)			
OR				
8.	Optical isomerism Optical isomerism arises from the substituent of asymmetric carbon atom in a polymer repeating unit. Example is given below  Polypropylene Depending on the special arrangement of these substituents different structures are possible for polymers. They are isotactic, syndiotactic and atactic structure. Examples for these three forms for PP are given below. Isotactic  Syndiotactic  Atactic  Geometrical isomerism If the repeating unit of a polymer containing carbon carbon double bond, then such repeating units can arrange differently with respect to that double bond. This can result in repeating units with same chemical structure but different spatial arrangement or isomeric structures. Examples for the same is given below  Cis 1,4 Poly(isoprene)  trans 1,4 Poly(isoprene)	3.5	7	7
9.	Hydrolysis Hydrolysis is the reaction of water molecules with amide, ester and acetal linkages	3.5	7	7

	present in polymers. This reaction can result in chain scission as well as polymer modifications. Examples for hydrolysis are given below. Amide + H₂O → Carboxylic acid + Amine Ester + H₂O → Carboxylic acid + Alcohol Acidolysis Acidolysis is the reaction of carboxylic acid molecules with groups like amide groups present in polymers. This reaction can result in chain scission. Example for hydrolysis are given below Amide + Carboxylic acid → Carboxylic acid-terminated polymer + Amide-terminated polymer	3.5		
OR				
10	Crosslinking Polymers can be modified using crosslinking reaction. This method is extensively used for rubbers to improve their mechanical and thermal properties. Rubbers are normally weak materials having lower TS, modulus and high temperature stability. Example is the crosslinking of NR with sulphur. Crosslinking can improve all these properties. It also improves solvent resistance. In plastics crosslinking reactions are widely used for preparing thermosetting plastics. Example Amines used to cure epoxy resins. It is also used to crosslink thermoplastics as well used for some special purpose applications. Example :- Benzoyl peroxide used for the crosslinking of LDPE. Inter penetrating network (IPN) Interpenetrating polymer networks (IPNs) are a combination of two or more polymers in networks where a partial interlacing on the molecular scale is present in the matrix. The two or more networks can be envisioned to be entangled in such a way that they are concatenated and cannot be pulled apart, but not bonded to each other by any chemical bond. Good blending or mixing of properties of two polymers can be achieved in this way. It can be prepared by immersing a lightly cross-linked polymer into another	3.5	7	7

	monomer followed by its polymerizaion after penetration of monomer into the crosslinkednetwork. Here the domain size of dispersed phase will be less and in mos cases can obtain a coconinous morphology.			
11.	Hydroxyl groups in polymers can undergo many chemical reactions. Polymers can be modified using these reactions. Polarity and other properties of polymers can be modified in this way. Classic example is the modification of polyvinyl alcohol. It is soluble in water and this property can be tuned using following reaction  Polymers contain carboxylic acid groups. These carboxylic acid groups can undergo reactions shown by simple aldehydes. Polymers can be modified using these reactions. Examples for such reactions are given below. 	3.5	7	7
OR				
12.	Blending Mixing of two or more polymers to produce a material with new properties or modifying the properties of an existing polymer.Example:- Blending of SBR and NR. Low abrasion resistance, poor oxidation resistance, low crack initiation resistance, ect... of NR can be improved by blending it with SBR. Blending can be used for modifying properties of a polymer, Combining properties of two polymers, Controlling price of a polymer product, Modifying processability of polymers. Polymer alloy Polymer alloys are class of polymer blends where the blend components are having high compatibility.Example: - Polystyrene/Styrene buadiene block copolymer blend. Composite Composite is a mixture of two chemicals with different physical and chemical	3		
		2	7	7

	properties. Polymer composite consists of a polymer matrix and dispersed solid filler/s.Fillers incorporated into polymer is having higher modulus than that of polymer. If the polymer is having very good interaction with filler then the filler can improve the physical properties of polymer.Example:- Carbon black incorporated SBR (carbon black filler reinforces SBR polymer matrix), fiber reinforced plastic, ect...	2		
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BLUE PRINT
Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	10	28	3	3	3	9	2	14	8	26
2	12	35	2	2	2	6	4	28	8	36
3	11	32	2	2	4	12	2	14	8	28
4	10	28	2	2	1	3	4	28	7	33
Total	43	123	9	9	10	30	12	84	30	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	5	4
Understanding	111	90
Applying	7	6
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis

Course : Polymer science

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	Remember	1	2
I.2	M2.01	Understand	1	2
I.3	M2.04	Understand	1	2
I.4	M4.05	Remember	1	2
I.5	M1.02	Remember	1	2
I.6	M1.04	Remember	1	2
I.7	M3.03	Understand	1	2
I.8	M4.05	Understand	1	2
I.9	M3.02	Remember	1	2
II.1	M1.03	Understand	3	7
II.2	M2.03	Understand	3	7
II.3	M2.02	Understand	3	7
II.4	M3.02	Understand	3	7
II.5	M3.01	Understand	3	7
II.6	M3.04	Understand	3	7
II.7	M4.02	Understand	3	7
II.8	M3.03	Understand	3	7
II.9	M1.03	Understand	3	7
II.10	M1.03	Understand	3	7
III.1	M1.04	Understand	7	17
III.2	M1.03	Understand	7	17
III.3	M2.04	Understand	7	17
III.4	M2.03	Apply	7	17
III.5	M2.04	Understand	7	17
III.6	M2.04	Understand	7	17
III.7	M3.04	Understand	7	17
III.8	M3.02	Understand	7	17

III.9	M4.03	Understand	7	17
III.10	M4.04	Understand	7	17
III.11	M4.02	Understand	7	17
III.12	M4.04	Understand	7	17

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

POLYMER SCIENCE
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define the term plastics with respect to Tg	M1.03	R
2	Define is crystallinity	M1.01	U
3	Give the equation for Mn	M2.03	U
4	Light scattering technique is used to determine-----	M2.04	R
5	Zeigler Natta catalyst is used for the production of----- polymers	M3.03	R
6	Give the structure of cis 1,4 Polyisoprene	M3.04	R
7	Name one physical methods of polymer modification	M4.04	U
8	What is copolymerization	M4.04	U
9	Define flow temperature	M1.02	R

PART B

V. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Explain geometric isomerism with example	M3.02	U
2	Distinguish between crystalline and amorphous polymers with examples	M1.03	U
3	Briefly explain GPC for determination of molecular weight	M2.04	U
4	Write short note on the determination of viscosity average molecular weight	M2.04	U
5	Explain molecular weight determination by end group analysis	M2.04	U
6	Discuss about the possible microstructures of polybutadiene	M3.04	U
7	Give brief explanation of microstructures based on chemical structure	M3.01	U
8	Write short note on TGA	M1.04	U

9	Explain the effect of hydroxyl groups	M4.02	Understand
10	Discuss the modification of polymers by composite preparation.	M4.04	Understand

PART C

VI. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Discuss the factors affecting crystallizability	M1.03	U
OR			
2	Explain the X-Ray diffraction method of finding crystallinity in polymers	M1.04	U
3	Explain the technique for determining molecular weight of polymers by ebulliometry	M2.04	U
OR			
4	In a particular sample of polymer 100 polymers have molecular weight 10^3 each. 200 molecules have molecular weight 10^4 each and 200 molecules have molecular weight 10^5 each. Calculate the number average and weight average molecular weights	M2.03	A
5	Derive the expressions for M_w	M2.03	U
OR			
6	What is a Ziegler-Natta catalyst. State its relevance with two examples. List any two combinations of Z-N catalyst.	M3.03	U
7	Differentiate optical isomerism and geometrical isomerism. Explain optical isomerism using the examples of isotactic, syndiotactic and atactic polypropylene	M3.02	U
OR			
8	Discuss possible microstructures of CR and polybutadiene	M3.04	U
9	Explain the polymer reactions such as hydrogenation, Addition and hydrolysis with examples	M4.03	U
OR			
10	Illustrate the effect of different functional groups in polymer	M4.02	U
11	Explain polymer modification by crosslinking and blending	M4.04	U
OR			
12	Explain polymer modifications by copolymerization and grafting	M4.04	U

Prepared By:

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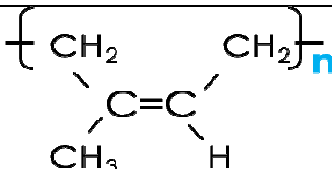
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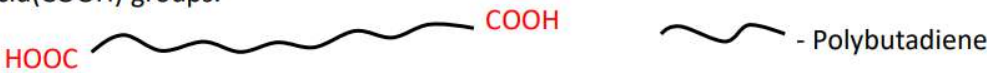
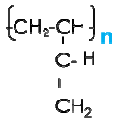
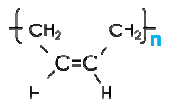
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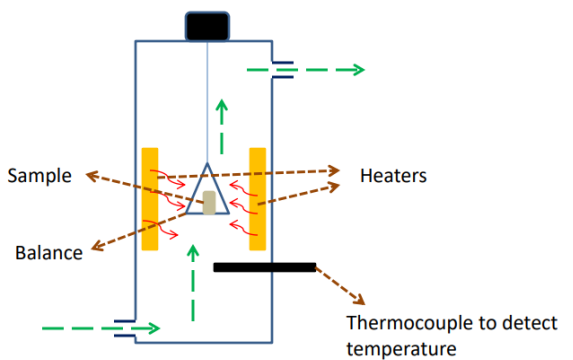
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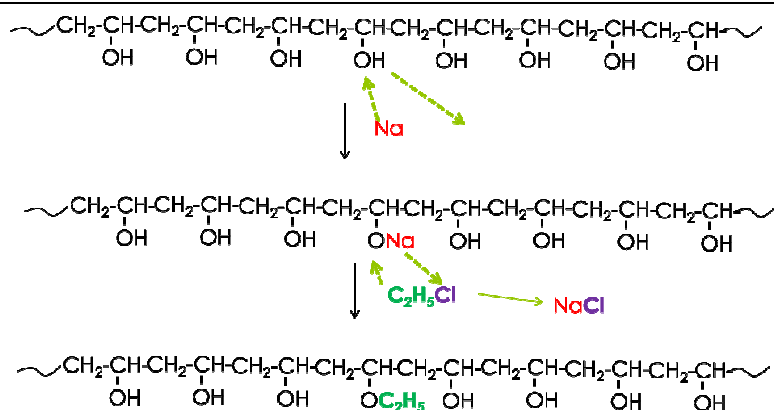
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SCHEME OF VALUATION**(Scoring Indicators)**

Revision : 2021		Course Code :		
Course Title :Polymer science- Set 2				
Qst. No	Scoring Indicator	Split up Score	Sub Total	Total
	PART A			
I. 1.	Plastics are polymers with Tg higher than room temperature	1	1	1
2.	Fraction of crystalline domains in a polymer sample	1	1	1
3.	Number average molecular weight ($\overline{M}_n$)$= \frac{\sum N_i M_i}{\sum N_i}$	1	1	1
4.	Weight average molecular weight	1	1	1
5.	Sterio regular	1	1	1
6.			1	1
7.	Blending, Alloying, composite preparation	1	1	1
8.	If the polymerization system containing more than one monomer that produce different repeating units, then such polymerization is called copolymerization	1	1	1
9.	Thetemperature at which molecular mobility sets and thepolymer passes from rubbery state into liquid state iscalled Flow temperature or flow point (Tf)	1	1	1
	PART B			
1.	Geometrical isomerism If the repeating unit of a polymer containing carbon carbon double bond, then such repeating units can arrange differently with respect to that double bond. This can result in repeating units with same chemical structure but different spacial arrangement or isomeric structures. Examples for the same is given below		3	3

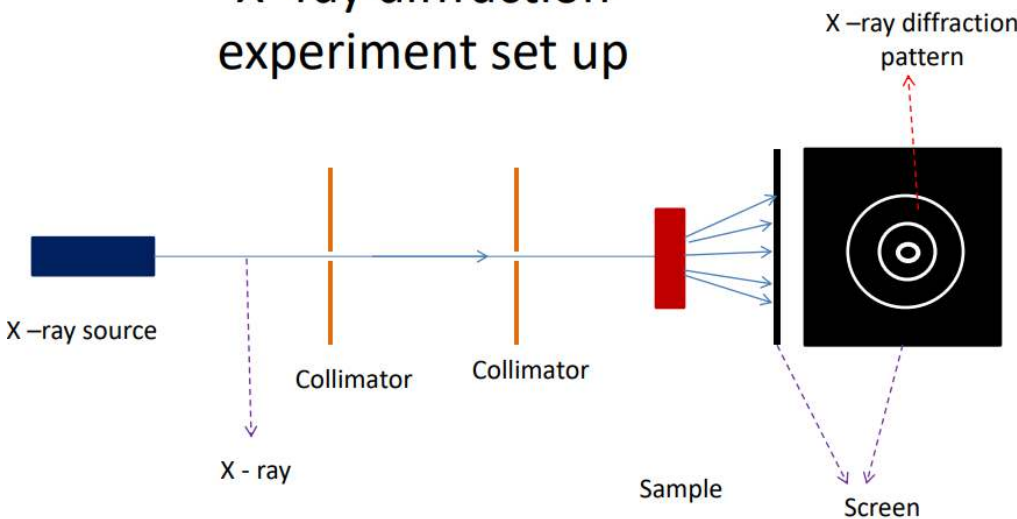
	● Rubber bulb is removed so that the liquid starts flowing through the capillary from top to bottom. ● Calculate the time required to lower the liquid level from M1 to M2 using a stop watch. ● Repeat the same for polymer solutions with different concentrations. ● Reduced viscosity value and inherent viscosity value for each concentration are calculated and plotted against concentration ● Intrinsic viscosity value is calculate from the graph ● Molecular weight is calculated using Mark – Houwink equation			
5.	End group analysis Useful to find the molecular weight of polymers having functional in their chain ends. In order to determine molecular weight molecule should contain reactive end groups. Number/concentration of end groups in a polymer sample can be found by titration method. Example: Determination of the molecular weight of carboxyl terminated polybutadiene. Here every polymer chain contains 2 carboxylic acid(COOH) groups.  1. Known amount CTPB is dissolved in toluene –ethanol mixture. 2. It is titrated against standardized KOH solution in alcohol. 3. Carboxyl equivalent = $\frac{V \cdot N}{W \cdot 1000}$ V – volume of KOH solution used N – Normality of KOH solution W – weight of sample 4. Number average molecular weight ($\overline{M}_n$) = $\frac{2}{\text{Carboxyl equivalent}}$	3	3	3
6.	Polybutadiene can have two different types of repeating units. It is because of difference in addition reaction of double bonds present in butadiene monomers. Two possible microstructures are given below.  1,2 poly(buadiene)  1,4 Poly(butadiene)	1*3	3	3

7.	Microstructure of the polymer can be studied based on chemical structure or chemical composition. Basic chemical structure and composition determines the properties of polymers. Based on the chemical structure polymers are classified as ● Organic and inorganic polymers -Organic polymers are polymers which contain carbon atoms in the polymer backbone. Eg:- polyethylene, polypropylene, PS ect...Inorganic polymers are polymers with carbon free backbone. Eg:- Silicone polymers, Polygermaneect.. ● Homo chain and hetero chain polymers – Polymer backbonecontaining only one type of atom is called homo chain polymers eg:- PE and others called hetero chain polymers eg:- nylon. ● Homo polymers and copolymers – Polymers made from same monomers are called homopolymerseg:- PP, Polymers with more than one type of repeaing units are called copolymers eg:- SBR.			
8.	 • Sample is placed on the balance before starting the experiment. • Sample is heated at a specific rate (Eg: - 1 °C per minute). • A gas (eg:- nitrogen, air, oxygen ect...) is passes through the sample chamber throughout the experiment. This helps in the removal of volatile materials evolving from sample. • Weight of the sample is measured as a function of temperature. • Weight of the sample is plotted against temperature. • When sample degrades a reduction in weight is observed. • From this the degradation temperature of a polymer can be found • Degradation temperature is different for different polymers			
9.	Hydroxyl groups in polymers can undergo many chemical reactions. Polymers can be modified using these reactions. Polarity and other properties of polymers can be modified in this way. Classic example is the modification of polyvinyl alcohol. It is soluble in water and this property can be tuned using following reaction	3	3	3



10.	Composite Composite is a mixture of two chemicals with different physical and chemical properties. Polymer composite consists of a polymer matrix and dispersed solid filler/s. Fillers incorporated into polymer is having higher modulus than that of polymer. If the polymer is having very good interaction with filler then the filler can improve the physical properties of polymer. Example:- Carbon black incorporated SBR (carbon black filler reinforces SBR polymer matrix), fiber reinforced plastic, etc...	3	3	3
III				
1.	Factors affecting crystallizability •Geometric regularity – Linear polymers can crystallize easily compared to branched polymer because linear polymers are having more regular structure. Example linear PE (HDPE) crystallize better than branched PE (LDPE) •Stereo regularity/Tacticity – If the polymer is more stereo regular then it can crystallize easily. Isoactic and syndiotactic polymers can crystallize easily because they are having structural regularity. •Macromolecular geometry – macro molecular geometry can affect crystallization. NR molecules having a coiled structure and they can crystallize upon cooling or stretching. Gutta-percha molecules are chemically identical but having a rod like structure. They cannot crystallize •Copolymerization – Copolymerization produce polymer with less structural regularity. So copolymerization reduce crystallizability. Example polyethylene and polypropylene are crystalline but their copolymer is amorphous. •Intermolecular forces - Inter molecular forces enhances close packing of polymer chains. So intermolecular forces favour crystallization. •Bulky side groups - Bulky side groups reduce polymer chain flexibility and make chain packing difficult. So they reduce crystallizability.	7	7	7

OR

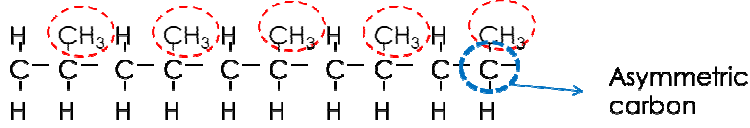
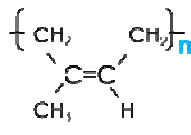
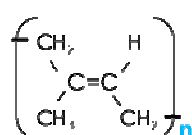
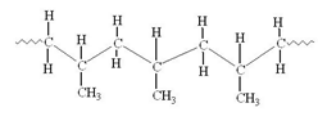
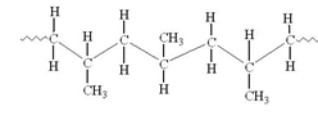
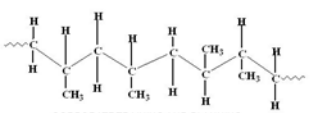
2.	X- ray diffraction experiment set up  The diagram illustrates the X-ray diffraction experiment setup. It shows an X-ray source on the left, emitting a beam that passes through two collimators (represented by vertical orange bars). The beam then passes through a sample (a red rectangle). From the sample, multiple diffracted beams emerge, which are captured by a screen on the right. The screen displays a characteristic X-ray diffraction pattern, consisting of concentric circles. Labels include: X-ray source, Collimator, Collimator, Sample, Screen, and X-ray diffraction pattern. X ray diffraction experiment set up is given in the figure. • X ray beam from a source passes through two collimators. This process helps in streamlining the X ray waves. • This X ray is then passed through the sample • X rays interact with the sample and emerge from the sample • Emerging X ray beams from the sample form a diffraction pattern on the detector. A sharp diffraction pattern is observed for crystalline materials.	7	7	7
3.	• Molecular weight determination of polymers using ebulliometry is based on the colligative property change of a pure solvent by the addition of solute • Boiling point of a liquid or a solvent increases by the addition of a solute. • In case of polymer solutions the boiling point of the pure solvent increases by the addition of polymer into it. Ebulliometry is working on this principle • Here polymer solutions with different polymer concentrations are prepared. • Boiling point of each solution is measured and the elevation in boiling is measured at each case • Elevation in boiling point is divided by concentration and plotted against concentration and molecular weight is found using the graph Relation between molecular weight of polymer sample and its concentration in solution is shown in the equation $\frac{\Delta T_b}{C} = \frac{RT_b^2}{\rho \Delta H_v} \left(\frac{1}{M_n} + BC \right)$ C – concentration of polymer in the solution ΔT_b – Difference between the boiling points of pure solvent and the solution T_b – Boiling point of pure solvent R – Universal gas constant ΔH_v – Heat of vapourization of solvent ρ – Density of solvent B – Constant	7	7	7

	$\left(\frac{\Delta T_f}{C}\right)_{C \rightarrow 0}$ found from the plot and the molecular weight is found using the equation given below $\left(\frac{\Delta T_b}{C}\right)_{C \rightarrow 0} = \frac{RT_b^2}{\rho \Delta H_v} * \frac{1}{M_n}$			
OR				
4.	$M_1 = 10^3, N_1 = 100$ $M_2 = 10^4, N_2 = 200$ $M_3 = 10^5, N_3 = 200$ Then $M_1N_1 = 100*10^3 = 10^5$ $N_1M_1^2 = N_1M_1 * M_1 = 10^5*10^3 = 10^8$ $M_2N_2 = 200*10^4 = 2*10^6$ $N_2M_2^2 = N_2M_2 * M_2 = 2*10^6*10^4 = 2*10^{10}$ $M_3N_3 = 200*10^5 = 2*10^7$ $N_3M_3^2 = N_3M_3 * M_3 = 2*10^7*10^5 = 2*10^{12}$ $M_n = (M_1N_1 + M_2N_2 + M_3N_3)/(N_1 + N_2 + N_3) = (10^5 + 2*10^6 + 2*10^7)/(100 + 200 + 200)$ $= 44200$ $M_w = (N_1M_1^2 + N_2M_2^2 + N_3M_3^2)/(M_1N_1 + M_2N_2 + M_3N_3)$ $= (10^8 + 2*10^{10} + 2*10^{12})/(10^5 + 2*10^6 + 2*10^7) = 91407$	7	7	7
5.	Let there be N_1 number of polymer molecules with M_1 molecular weight N_2 number of polymer molecules with M_2 molecular weight N_3 number of polymer molecules with M_3 molecular weight Ect... N_i number of polymer molecules with M_i molecular weight Weight of polymer molecules with molecular weight M_1 = Number of molecules with molecular weight M_1 * Molecular weight (M_1) = $N_1 * M_1$ = $N_1 M_1$	3.5	7	7
		3.5		

	Weight of polymer molecules with molecular weight M_2 = $N_2 M_2$... Weight of polymer molecules with molecular weight M_i = $N_i M_i$ Total weight of polymer sample = $N_1 M_1 + N_2 M_2 + N_3 M_3 + \dots + N_i M_i$ = $\sum N_i M_i$ Weight fraction of polymer molecules with molecular weight M_1 = $\frac{\text{Weight of polymer molecules with } M_1 \text{ Molecular weight}}{\text{Total weight of all polymer molecules}}$ = $\frac{N_1 M_1}{\sum N_i M_i}$ Weight fraction of polymer molecules with molecular weight M_2 = $\frac{N_2 M_2}{\sum N_i M_i}$ Weight fraction of polymer molecules with molecular weight M_i = $\frac{N_i M_i}{\sum N_i M_i}$			
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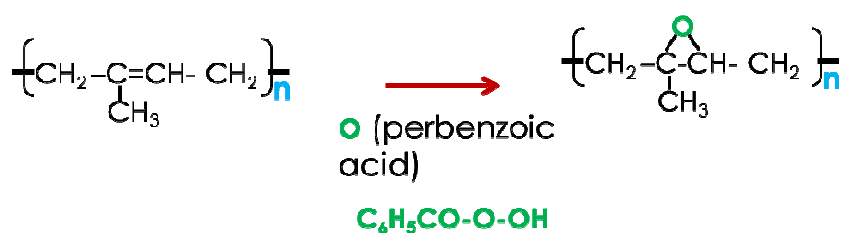
OR

6..	Zeigler natta catalysts are coordination catalysts. A zeiglernatta catalyst consists of two components, Halides of transition metals (Catalyst) and Alkyls, aryls and hydrides of group 1 to 4 elements (Co catalyst). Catalyst then forms coordination complex with cocatalyst. Here the mechanism is explained by taking the example of titanium chloride aluminum alkyl combination. When catalyst mixed with co catalyst the second component gets chemisorbed onto the solid surface of first component. This results in the formation of an electron deficient bridge complex It is used for the production of stereo regular polymers. Used in the production of polypropylene and isoprene rubber.			
-----	--	--	--	--

	Examples:- Titanium tetrachloride + triethyl aluminium, Titaniumtrichloride + triethyl aluminium			
7..	Optical isomerism arises from the substituent of asymmetric carbon atom in a polymer repeating unit. Example is given below  Polypropylene Geometric isomerism is observed in polymers with repeating units having double bond. Here the substituents on C=C carbon atoms should be different. Example is given below  Cis 1,4 Poly(isoprene)  trans 1,4 Poly(isoprene) 1,4 addition Depending on the special arrangement of these substituents different structures are possible for polymers. They are isotactic, syndiotactic and atactic structure. Examples for these three forms for PP are given below. Isotactic  Syndiotactic  Atactic 			
OR				
8.	Different structures and spatial arrangements are possible for the repeating units of CR and PBD depending on the reaction of monomers and their special arrangement during polymerization. Possible structures of CR and PBD are given below.	7	7	7

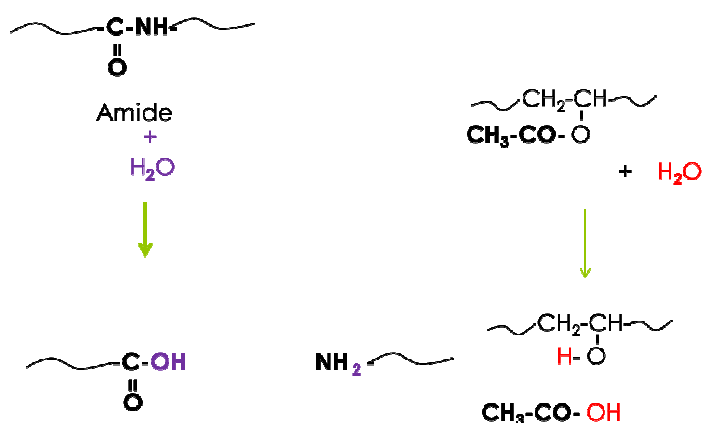
	$\begin{array}{c} \text{CH}_2-\text{CH} \\ \\ \text{C}-\text{Cl} \\ \\ \text{CH}_2 \end{array} \quad n$ 3,4 poly(chloroprene) 3,4 addition $\begin{array}{c} \text{CH}_2 \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{CH}_3 \quad \text{H} \end{array} \quad n$ Cis 1,4 Poly(chloroprene) 1,4 addition $\begin{array}{c} \text{Cl} \\ \\ \text{CH}_2-\text{C} \\ \\ \text{CH} \\ \\ \text{CH}_2 \end{array} \quad n$ 1,2 poly(chloroprene) 1,2 addition $\begin{array}{c} \text{CH}_2 \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{Cl} \quad \text{CH}_2 \end{array} \quad n$ trans 1,4 Poly(chloroprene) $\begin{array}{c} \text{CH}_2-\text{CH} \\ \\ \text{C}-\text{H} \\ \\ \text{CH}_3 \end{array} \quad n$ 1,2 poly(buadiene) $\begin{array}{c} \text{CH}_2 \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array} \quad n$ 1,4 Poly(butadiene)			
9.	Hydrogenation is the process of adding hydrogen into an unsaturated molecule. Polymers with C=C can be modified using hydrogenation. Properties like oxidation resistance can be modified using this. Examples of hydrogenation are given below. Polystyrene Ni catalyst Pressure 200°C H₂ Addition process is a chemical reaction of adding an element like oxygen into a polymer that contains reactive sites like C = C. Example of an addition reaction is given below.			

Oxygen atoms can be added into polyisoprene using the following reaction



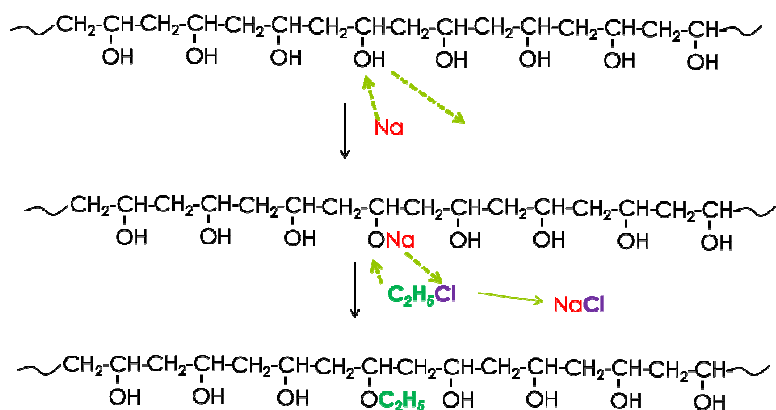
Hydrolysis

Hydrolysis is the reaction of water molecules with amide, ester and acetal linkages present in polymers. This reaction can result in chain scission as well as polymer modifications. Examples for hydrolysis are given below.

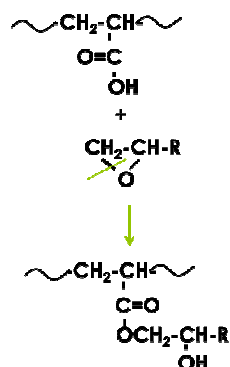
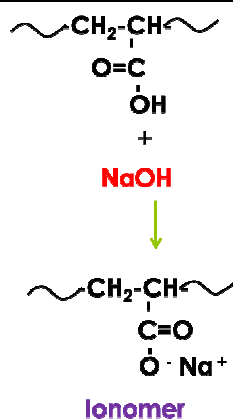


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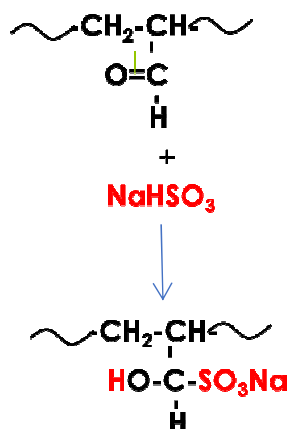
- 10.. Hydroxyl groups in polymers can undergo many chemical reactions. Polymers can be modified using these reactions. Polarity and other properties of polymers can be modified in this way. Classic example is the modification of polyvinyl alcohol. It is soluble in water and this property can be tuned using following reaction



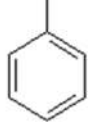
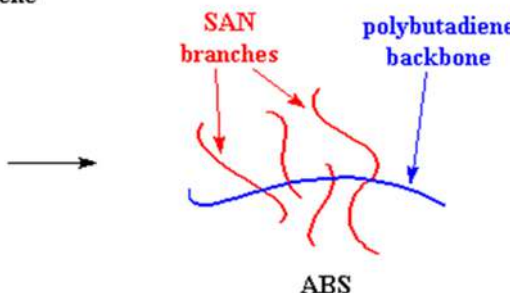
Polymers contain carboxylic acid groups. These carboxylic acid groups can undergo reactions shown by simple aldehydes. Polymers can be modified using these reactions. Examples for such reactions are given below.



Some of the polymers contain aldehyde groups. These aldehyde groups can undergo reactions shown by simple aldehydes. Example for the polymer containing aldehyde group is polyacrolein. Aldehyde group in polyacrolein can react with sodium bisulfate. This reaction can convert polyacrolein from a water insoluble polymer into a water soluble polymer.



11..	Blending Mixing of two or more polymers to produce a material with new properties or modifying the properties of an existing polymer. Example:- Blending of SBR and NR. Low abrasion resistance, poor oxidation resistance, low crack initiation resistance, ect... of NR can be improved by blending it with SBR. Blending can be used for modifying properties of a polymer, Combining properties of two polymers, Controlling price of a polymer product, Modifying processability of polymers. Crosslinking Polymers can be modified using crosslinking reaction. This method is extensively used for rubbers to improve their mechanical and thermal properties. Rubbers are normally weak materials having lower TS, modulus and high temperature stability. Example is the crosslinking of NR with sulphur. Crosslinking can improve all these properties. It also improves solvent resistance. In plastics crosslinking reactions are widely used for preparing thermosetting plastics. Example Amines used to cure epoxy resins. It is also used to	3.5		
		3.5	7	7

	crosslink thermoplastics as well used for some special purpose applications. Example :- Benzoyl peroxide used for the crosslinking of LDPE.			
OR				
12.	Copolymerization Copolymers generally show a combination of properties of homo polymers of monomers used. Mechanical property of a polymer depends on the types of monomers used and their ratio in copolymer. For example copolymer of styrene and butadiene. Homo polymer of styrene is a brittle plastic with high tensile strength and modulus whereas homo polymer of butadiene is a weak rubbery material with lower modulus and tensile strength. If styrene content in copolymers is high then that copolymer will show higher modulus and higher tensile strength compared to the copolymer with low styrene content. High styrene copolymer is leathery whereas low styrene copolymer is a rubber. Copolymerization reduces chances of crystallization. It is observed in copolymerization of ethylene and propylene. In such cases copolymerization is reducing modulus and tensile strength to a considerable extent. Grafting reaction Grafting is the process of attaching a branch of different polymer onto an existing polymer. It is possible to modify polymers using this technique. It can be done by mixing two polymers in presence of a suitable chemical or by mixing a monomer with another polymer followed by its polymerization. Example for graft copolymer is ABS (Acrylonitrile butadiene styrene copolymer) produced by mixing polybutadiene with styrene and acrylonitrile. $\text{CH}_2=\text{CH}_2 + \text{CH}_2=\text{CH}_2 + \text{---}\text{CH}_2\text{---CH=CH---CH}_2\text{---}_n$  styrene $\text{C}\equiv\text{N}$ acrylonitrile polybutadiene 	3.5		
			7	7

BLUE PRINT

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	10	28	3	3	2	6	2	14	7	23
2	12	35	2	2	3	9	3	21	8	32
3	11	32	2	2	3	9	3	21	8	32
4	10	28	2	2	2	6	4	28	8	36
Total	45	123	9	9	10	30	12	84	30	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	5	4
Understanding	111	90
Applying	7	6
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.03	Remember	1	2
I.2	M1.01	Understand	1	2
I.3	M2.03	Understand	1	2
I.4	M2.04	Remember	1	2
I.5	M3.03	Remember	1	2
I.6	M3.04	Remember	1	2
I.7	M4.04	Understand	1	2
I.8	M4.04	Understand	1	2
I.9	M1.02	Remember	1	2
II.1	M3.02	Understand	3	7
II.2	M1.03	Understand	3	7
II.3	M2.04	Understand	3	7
II.4	M2.04	Understand	3	7
II.5	M2.04	Understand	3	7
II.6	M3.04	Understand	3	7
II.7	M3.01	Understand	3	7
II.8	M1.04	Understand	3	7
II.9	M4.02	Understand	3	7
II.10	M4.04	Understand	3	7
III.1	M1.03	Understand	7	17
III.2	M1.04	Understand	7	17
III.3	M2.04	Understand	7	17
III.4	M2.03	Apply	7	17
III.5	M2.03	Understand	7	17
III.6	M3.03	Understand	7	17
III.7	M3.02	Understand	7	17
III.8	M3.04	Understand	7	17
III.9	M4.03	Understand	7	17
III.10	M4.02	Understand	7	17
III.11	M4.04	Understand	7	17
III.12	M4.04	Understand	7	17

